Characterization and evidence: Exercises

> Reference: Jennifer Rupp, Daniele Ielmini, Ilia Valov (eds.), Resistive Switching: Oxide Materials, Mechanisms, Devices and Operations, Springer, 2022. DOI: 10.1007/978-3-030-42424-4.

Numerical examples are hypothetical.

E01 Time resolution

An ideal instantaneous image sequence starts with frame 1 at $t = 0$ and has a 30 ms interval. A feature is absent in frame 3 and present in frame 4. Assume it becomes detectable once and remains detectable. Bound its first detectable appearance and give two experimental limitations omitted here.

E02 Tunneling gap

Under $I \propto V \rho_{\mathrm{eff}} e^{-2\kappa z}$, assume V, ρ_{eff} and $\kappa = 10\,\mathrm{nm}^{-1}$ remain fixed as z increases by 0.10 nm. Calculate I_2/I_1 and explain why a constant-current topograph is not pure geometry.

E03 Gaussian width

Assume a one-dimensional linear shift-invariant response: both true profile and calibrated instrument response are Gaussian. The observed convolution has FWHM 10 nm and instrument FWHM is 6 nm. Estimate the underlying FWHM and give two circumstances where this estimate does not apply.

E04 Slice interpolation

At one registered position, current maps show 2, 6 and 10 nA at depths 0, 2 and 4 nm. Using linear interpolation, estimate currents at 1 and 3 nm and the depth where current is 7 nA. Is that depth a direct observation?

E05 Pressure modulation

At constant device current $1\,\mu$A, pressure changes conductance from $10\,\mu$S to $12\,\mu$S; assume $G = I/V$ at the read point. Compute resistances, voltages and fractional conductance change. Must pressure lower resistance in every device?

E06 Measurement design

An RRAM I–V curve shows hysteresis, and a CAFM map contains a local current hotspot. Propose three controls or complementary measurements to distinguish conductive-path change, contact change and measurement disturbance. What can one hotspot not establish directly?

E07 In situ imaging and chemical evidence

A bright feature appears after SET. Why can brightness alone not identify an element? If synchronized I–V, EDS/EELS and an image after RESET are available, how would you combine them and which perturbations would you check?

E08 Reconstruction limits

Explain how depth calibration, image registration, interpolation and current threshold affect a 3D slice reconstruction. Why is this not a real-time 3D video of an undisturbed device?

Characterization and evidence: Exercises (continued)

E09 Unresolved feature

In the Gaussian model from E03, the instrument FWHM is 6.0 $\pm$ 0.5 nm and observed FWHM is 5.8 $\pm$ 0.5 nm. Is a positive sample width established? Explain what should be reported instead.

E10 Compare CAFM configurations

One researcher programs an intact cell and removes its top electrode before a CAFM scan. Another writes a bare film using the CAFM tip as electrode. Compare electrical boundaries and two possible artifacts; why need their maps not coincide?

E11 Retention or scan artifact?

After a high-bias STM write at one location, a low-bias current map differs from baseline. Design a write–read–wait–read experiment with an unprogrammed control and one tip-related control. State a supported conclusion if the change persists and one unjustified conclusion.

E12 Coherent double-barrier model

Assume one spin-degenerate channel at zero temperature and infinitesimal bias: $G = G_Q T(\phi)$, $G_Q = 2e^2/h$, and $T(\phi) = [1 + 3\sin^2(\phi/2)]^{-1}$. Suppose compression increases ϕ by $\pi/6$ while barriers remain fixed. Find T before and after, and $\Delta G/G$, for initial phase $\pi/2$ and separately $3\pi/2$. Name two limits of this model.

E13 EDS evidence and limitations

For an invented two-state example, the Ag-peak window contains 80 counts after SET and 50 after RESET. In each, an estimated background of 20 counts is subtracted. Assume equal exposure and collection geometry. Calculate both net counts and their ratio. Does this ratio alone establish an Ag concentration ratio or a unique switching mechanism? Give two timing or sample controls.

E14 Thresholded slice area

A hypothetical registered slice has a 2×3 current grid in nA: $[3, 4, 8]$ and $[0, 6, 2]$. Every cell covers $10\,\mathrm{nm} \times 10\,\mathrm{nm}$. Compute the displayed area when cells with current at least 4 nA are selected, and again for at least 7 nA. Is either area necessarily the physical filament cross-section? Explain the role of registration and tip changes.

E15 Growth model and state-dependent evidence

Within the qualitative model discussed in this chapter, suppose mobile-ion transport is high and redox is slow. Where is nucleation favored, and what growth morphology is expected? A separate Pt/TiO_2/Pt specimen has higher resistance after RESET and loses a diffraction reflection attributed to Ti_4O_7; another Ta-oxide device shows different STEM-EELS spectra in HRS and LRS. What does each observation support? Give one control and one alternative or limitation for each, and explain whether contrast alone establishes a unique phase.